Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

assumption: a condition accepted for using a model

01 Bring back an earlier idea

A 3 kg block slides down a 30° ramp with 8.7 N friction up the ramp.
Use g=9.8 m/s².
Find along-ramp acceleration and normal force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = \Sigma F/m$; $v_f = v_i + a\Delta t$.
For a constant-acceleration interval. Check signs, units and assumptions; keep the original step visible when revising.

Worked reasoning: Claim: a car moving east must accelerate east. Counterexample: it moves east while braking, so acceleration points west. A single valid counterexample can disprove a universal claim.

Your turn: Draw velocity and acceleration arrows for that counterexample. Write a corrected statement about their directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Find the first bad step

An object starts at x=5 m, then reaches 1 m after 2 s. Student writes v=1/2=.5 m/s. Correct it.
Explain when dividing final position by time would work.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

04 Check the axis

A 10 N vector is 30° west of north. Student writes Fx=10 cos 30 and Fy=10 sin 30. Correct signs and trig choices.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Fresh recheck

Resolve 8 Nat 30° south of east. Then find a of a 5 kg object with 7 N right and 22 N left.
State whether the force information alone establishes the motion direction.

Word help: resolve = separate a vector into components

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

Velocity changes from -5 to -35 m/s in 5 s, then stays at -35 m/s for 9 s.
Find acceleration and describe the speed trend in each interval.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

For $x_0=3\pm1$ m, $v=4\pm.5$ m/s, $t=2$ s, find central position and interval.
Identify one earlier error pattern you checked before finishing.

Sentence starter: The step I would change is ... . The evidence is ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A student calculates acceleration using the applied force alone and gets 5 m/s² for a 2 kg cart. The measured acceleration is 3 m/s². Infer a possible opposing force, then name another explanation to test.

Word help: infer = use evidence to reach a conclusion

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3